Hamilton Sundstrand

A United Technologies Company

Date May 29, 2002
Time 12:20:01 PM

Engine	APS3200R	Test	APS3200R	Serial Number	1364	Assembly Number	4500001B
Test Run Type	REPAIR ATP	Operator	R.L. Rogers	Q.C Inspector	<	>	
Revision	46 A	Work Order	01423412	Test Document	Engine Manual Document A320 EM,49-20-00	Rev	25

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PARAGRAPH		-03, Para. 1	-03, Para. 2	-03, Para. 2	-04, Para. 2	-04, Para. 4	-04, Para. 7	-04, Para. 8	-04, Para. 9	-04, Para. 9	-04, Para. 9
SECTION		LEAK CHK	D/OIL CHK	OWN CHK	NTRL CHK	ECS MODE	SHAFT LD	NO LOAD	LEAK CHK	TIME CHK	INAL SAVE
DATAPOINT		B	D1	D2	H	I	L	M	N1	N2	SD
DATE		4/12/2002	4/12/2002	4/12/2002	4/12/2002	4/12/2002	4/12/2002	4/12/2002	4/12/2002	4/12/2002	4/12/2002
TIME		19:03	20:43	20:44	20:50	20:56	20:58	21:01	22:15	22:17	22:17
TEST CELL		7	7	7	7	7	7	7	7	7	7

Italics denotes SET Parameter

	Parameter	Units										
1	TIME.RUN	Min	10.07	6.13		3.38	5.17	3.14	2.63	34.31		
2	TOT_TIME	Sec	673.16	1096.69		1372.27	1682.27	1870.77	2028.28	4206.87		4271.58
3	STARTS	...	1.00	2.00		3.00	3.00	3.00	3.00	5.00		5.00
4	MAX.OIL	Psig	61.41	61.67		61.67	61.67	61.67	61.67	61.67		61.67
5	PENG.OIL	Psig	60.22	60.54		61.09	60.97	60.43	60.38			
6	T.GB.OIL	Deg.F	160.78	166.02		168.29	178.68	182.29	169.61	***0.00***		
	T.GENOIL	Deg.F	163.40	174.20		174.20	192.20	195.80	174.20	163.40		
	TIME.RD	Sec			57.40					0.00	57.20	
9	ET.SIP	Sec	43.85	43.52		41.99				42.35		
10	MAX.EGT	Deg.F	1094.86	1026.22	0.00	1082.48				1072.43	0.00	
11	RVD.ENG	in/sec	0.20	0.25		0.21	0.20	0.19	0.22	0.24		
12	RVD.GB	in/sec	0.20	0.19		0.15	0.13	0.11	0.15	0.17		
13	ENG.SPD1	Hz	19719.23	19719.23		19719.23	19719.23	19716.71	19716.71	19716.71		
14	ENG.SPD2	Hz	19725.86	19725.86		19725.86	19725.86	19725.86	19725.86	19723.34		
15	N.ENG	RPM	49298.07	49298.07		49298.07	49298.07	49291.78	49291.78	49291.78		
16	%ENG.SPD	%	100.00	100.00		100.00	100.00	99.98	99.98	99.98		
17	IGV.POS	Deg	82.00	82.00		-4.88	32.52	82.00	82.00	82.00		
18	BCV.LVDT	Vac	50.00	50.00		31.25	43.75	50.00	50.00	50.00		
19	P.BARO	In.Hg	29.53	29.56		29.56	29.56	29.57	29.57	29.58		
20	P.BARO.A	Psia	14.51	14.52		14.52	14.52	14.52	14.53	14.53		
21	P.INLET	In.H2O	-0.24	-0.26		-0.34	-0.31	-0.22	-0.24	-0.24		
22	TIN/1	Deg.F	71.07	69.69		72.68	73.32	73.10	72.07	71.13		
23	TIN/2	Deg.F	70.90	69.52		71.21	72.50	72.28	71.91	70.97		
24	TIN/3	Deg.F	70.61	70.11		71.79	72.43	72.43	72.27	70.89		
25	TIN/4	Deg.F	70.93	69.99		71.45	72.96	73.39	72.58	71.21		
26	TIN/5	Deg.F	71.16	71.09		71.68	74.50	74.72	73.04	71.66		
27	TIN/6	Deg.F	72.52	72.01		74.14	76.53	75.66	74.84	72.58		
28	TIN/7	Deg.F	72.28	71.78		72.80	74.09	74.53	73.72	72.34		
29	TIN/8	Deg.F	72.22	71.72		75.59	76.66	76.00	74.97	72.72		
30	TIN/9	Deg.F	71.95	71.01		72.47	73.54	75.29	74.70	72.45		
31	T.INLET	Deg.F	71.52	70.77		72.65	74.06	74.15	73.35	71.77		
32	T.AMB	Deg.F	73.29	72.35		74.25	75.33	75.11	75.17	74.23		
	LCIT	Deg.F	78.80	77.00		78.80	78.80	78.80	78.80	78.80		
	TIN.DEL	Deg.F	7.28	6.23		6.15	6.54	4.65	5.45	7.03		
35	DP/P	Psid	0.28	0.28		0.21	0.23	0.29	0.29	0.28		
36	P.EXDUCT	In.H2O	-1.71	-1.07		-1.57	-2.46	-1.36	-1.46	-1.61		
37	TEX/1	Deg.F	701.57	749.26		968.13	928.05	792.96	704.84	703.72		
38	TEX/2	Deg.F	685.71	733.66		979.97	930.81	780.38	690.66	688.91		

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DATAPOINT		B	D1	D2	H	I	L	M	N1	N2	SD
DATE		4/12/2002	4/12/2002	4/12/2002	4/12/2002	4/12/2002	4/12/2002	4/12/2002	4/12/2002	4/12/2002	4/12/2002
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TEST CELL		7	7	7	7	7	7	7	7	7	7

Italics denotes SET Parameter

	Parameter	Units										
39	TEX/3	Deg.F	679.02	725.92		1003.52	957.67	770.43	683.96	681.59		
40	TEX/4	Deg.F	677.63	736.06		1036.63	985.82	789.81	681.32	679.36		
41	TEX/5	Deg.F	688.72	754.60		996.97	965.16	806.60	692.20	690.67		
42	TEX/6	Deg.F	703.19	763.72		967.03	938.31	812.62	704.99	704.93		
43	TEX/AVG	Deg.F	689.31	743.87		992.04	950.97	792.13	692.99	691.53		
44	ETC/1	Deg.F	704.84	767.84		1051.64	1005.79	830.85	707.27	708.26		
	ETC/2	Deg.F	728.49	782.75		989.26	957.65	827.49	733.22	731.69		
	ETC/AVG	Deg.F	716.66	775.30		1020.45	981.72	829.17	720.25	719.97		
47	EGTCOR	Deg.F	702.40	756.10		1229.18	1123.65	801.03	701.64	705.15		
48	PFUEL.IN	Psig	35.67	33.88		22.57	25.57	32.93	35.98	35.79		
49	T.FUEL	Deg.F	67.62	66.89		69.01	68.77	68.34	68.40	69.43		
50	W.FUEL	PPH	164.68	188.21		279.01	262.03	203.39	164.78	164.30		
51	WFCOR	CORPP				274.54	268.24	205.79	167.19	166.65		
52	P.EXIT	Psig				46.42	35.28	-0.09	-0.06	1.89		
53	P.EXIT.A	Psia				60.94	49.80	14.43	14.46	16.42		
54	PBCOR	Psia				53.75	47.40			0.00		
55	T.EXIT	Deg.F				418.53	376.50	320.85	256.75	113.71		
56	TBCOR	Deg.F				479.48	408.71			32.00		
57	TBLEED	Deg.F			58.41	367.26	347.24	325.92	302.20	76.11		
58	P.BAIR/1	Psia				54.63	43.48	14.43	14.47	16.41		
59	PBAIRDIF	In.H2O				158.87	156.54	-0.37	-0.37	-0.31		
60	W.BLEED	ppm				167.30	148.62	0.00	0.00	0.00		
61	CW.BLEED	ppm				52.49	55.67	0.00	0.00	0.00		
62	WBCOR	CORPP				141.99	139.15	0.00	0.00	0.00		
63	VOLT.AVG	Vac		113.13		113.23	113.00	113.01	113.89	113.87		
64	AMP.AVG	Amps.ac		156.24		135.58	212.64	251.12	0.08	0.08		
65	FREQ	Hz		400.51		400.52	400.49	400.51	400.64	400.55		
66	WATTS	Kw		53.03		46.05	72.09	85.13	0.03	0.03		
67	KW.SHFT	Kw		60.98		53.38	82.18	97.08	5.45	5.45		
68	CKW.SHFT	Kw		61.75		54.07	83.23	98.27	5.51	5.51		
69	PCD	Psig	84.23	86.28	-0.05	91.63	90.40	86.90	83.84	83.91	-0.04	-0.04
70	%DO.CLS.SP	%		75.93								10.00
	%DO.OPN.SP	%		64.30								100.00

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DATAPOINT	B	D1	D2	H	I	L	M	N1	N2	SD
DATE	4/12/2002	4/12/2002	4/12/2002	4/12/2002	4/12/2002	4/12/2002	4/12/2002	4/12/2002	4/12/2002	4/12/2002
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TEST CELL	7	7	7	7	7	7	7	7	7	7

Fuel Type
ASTM D 1655 JET A

Operator Input

OIL SPEC	ETO 2389
ECB PART NUMBER	4500003H
SERIAL NUMBER	0002
SOFTWARE VERSION	4.1
REPORTED ACCUM HOURS	5230
REPORTED ACCUM CYCLES	6403

Test Information

TEST SPECIFICATION : APIC Engine Manual Document A320 EM

DATA REDUCTION SOFTWARE : APS3KCOR.CPP Version 1.0

Accumulate Parameters

			Seconds	Hours	Min	Sec
		RUN TIME THIS TEST	4271.578	1	11	12
TOTAL FUEL	210.49 LBS	TOTAL TIME	4271.578	1	11	12

Parameter Defintion

1	TIME.RUN	Time, This Run
2	TOT_TIME	Total Run Time.
3	STARTS	Number of Starts During Test.
4	MAX.OIL	Maximum Measured Oil Pressure during test.
5	PENG.OIL	Engine Oil Pressure.
6	T.GB.OIL	Engine Oil Temperature.
7	T.GENOIL	Generator Oil Temperature (from ECB).
8	TIME.RD	Run Down Time
9	ET.SIP	Elapsed Time for Start In Progress.
10	MAX.EGT	Maximum Exhaust Gas Temperature During Start Up (TEX/AVG)
11	RVD.ENG	Powerhead Vibration (40 to 1.2kHz).
12	RVD.GB	Gearbox Vibration (10 to 1.2kHz).
13	ENG.SPD1	Engine Speed Sensor No.1.
14	ENG.SPD2	Engine Speed Sensor No.2.
15	N.ENG	Engine Speed

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TEST CELL	7	7	7	7	7	7	7	7	7	7

16	%ENG.SPD	Percent Engine Speed
17	IGV.POS	Inlet Guide Vane Position.
18	BCV.LVDT	Bleed Control Valve Position.
19	P.BARO	Corrected Barometric Pressure
20	P.BARO.A	Corrected Barometric Pressure
21	P.INLET	Engine Inlet Pressure.
22	TIN/1	Compressor Inlet Temperature (individual probe reading)
3	TIN/2	Compressor Inlet Temperature (individual probe reading)
24	TIN/3	Compressor Inlet Temperature (individual probe reading)
25	TIN/4	Compressor Inlet Temperature (individual probe reading)
26	TIN/5	Compressor Inlet Temperature (individual probe reading)
27	TIN/6	Compressor Inlet Temperature (individual probe reading)
28	TIN/7	Compressor Inlet Temperature (individual probe reading)
29	TIN/8	Compressor Inlet Temperature (individual probe reading)
30	TIN/9	Compressor Inlet Temperature (individual probe reading)
31	T.INLET	Compressor Inlet Temperature (average)
32	T.AMB	Ambient Cell Air Temperature
33	LCIT	Load Compressor Inlet Temperature (from ECB).
34	TIN.DEL	Inlet Temp difference between LCIT and average of Cell probes
35	DP/P	APU DeltaP/P Sensor.
36	P.EXDUCT	Exhaust Duct Pressure
37	TEX/1	Exhaust Gas Temperature (Individual Rake Probe)
38	TEX/2	Exhaust Gas Temperature (Individual Rake Probe)
39	TEX/3	Exhaust Gas Temperature (Individual Rake Probe)
40	TEX/4	Exhaust Gas Temperature (Individual Rake Probe)
41	TEX/5	Exhaust Gas Temperature (Individual Rake Probe)
42	TEX/6	Exhaust Gas Temperature (Individual Rake Probe)
43	TEX/AVG	Exhaust Gas Temperature (rake average)
44	ETC/1	Engine Thermocouple No.1
45	ETC/2	Engine Thermocouple No.2
46	ETC/AVG	Engine Exhaust Thermocouple (average).
47	EGTCOR	Corrected Exhaust Gas Temperature (engine probe average)
'8	PFUEL.IN	Regulated Fuel Boost Pressure
)	T.FUEL	Fuel Temperature
50	W.FUEL	Mass Fuel Flow
51	WFCOR	Corrected Mass Fuel Flow
52	P.EXIT	Bleed Air Exit Pressure.
53	P.EXIT.A	Bleed Air Exit Pressure.

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TEST CELL	7	7	7	7	7	7	7	7	7	7

54	PBCOR	Corrected Bleed Air Exit Pressure.
55	T.EXIT	Bleed Air Exit Temperature.
56	TBCOR	Corrected Bleed Air Exit Temperature.
57	TBLEED	ASME Orifice Run, Air Temperature
58	P.BAIR/1	ASME Orifice Run, Air Pressure [P.BAIR + (P.BARO/2.036)]
59	PBAIRDIF	ASME Orifice Run, Air Static Delta P
60	W.BLEED	ASME Orifice Run, Air Flow.
1	CW.BLEED	Bleed Air Flow (Corrected at Bleed Scroll Outlet).
2	WBCOR	Corrected Bleed Air Flow.
63	VOLT.AVG	Voltage Output (3-phase average)
64	AMP.AVG	Current Output (3-phase average)
65	FREQ	Generator Frequency
66	WATTS	Generator Load.
67	KW.SHFT	Calculated Shaft Load.
68	CKW.SHFT	Corrected Shaft Load.
69	PCD	Compressor Discharge Pressure
70	%DO.CLS.SP	Percent engine speed when De-Oiling valve closed
71	%DO.OPN.SP	Percent engine speed when De-Oiling valve opened